

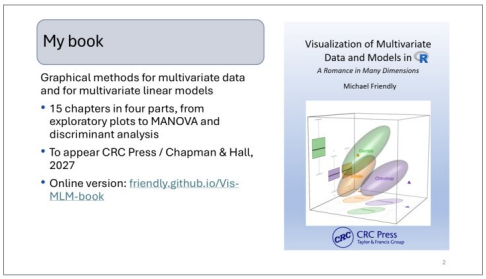
1. Visualizing Multivariate Data and Models in R

I'm so glad to be with you here today.

I just returned from Leuven, where I was on the jury for the Rousseeuw Prize in Statistics, which was happily awarded to The R Project in Statistical Computing & Graphics

As a bonus, daughter's family lives in Utrecht, so I was happy to see them all.

And now, I'll give the first public talk about my current book.



2. My book

This book is the outcome of what, in one way or another, I've been working on for nearly my entire career. I'm happy to be able to share it with you today

It started in my first grad course, Multivariate Data Analysis when I arrived at York in 1971

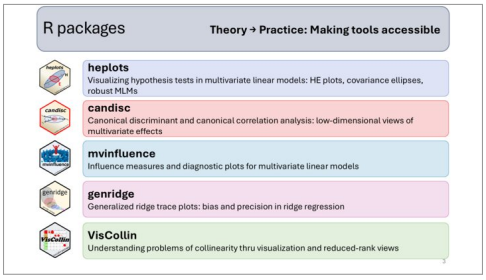
How to teach the essential ideas of multivariate analysis?

– Notation: Matrix algebra?

- Visual understanding: Geometry

How to make it easier for researchers to use these methods to answer multivariate questions?

The print copy is in the hand of Copy Editing Surgeons at CRC press.

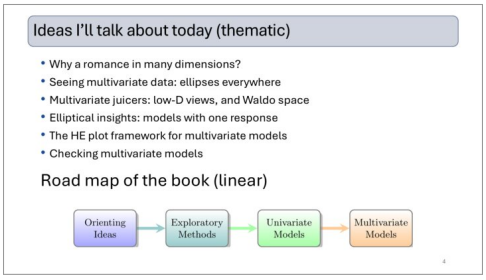


3. R packages

A large part of my work reflects Chairman Mao's assertion of Theory into Practice, and Tukey's collorary that statistical / graphical methods are only useful if people can use them.

In the work towards this book, I created a number of R packages to make these new graphical methods easily accessible.

These five packages implement most of what follows: heplots (HE plots and related methods), candisc (canonical views), mvinfluence (multivariate influence), genridge (ridge trace plots).



4. Ideas I'll talk about today (thematic)

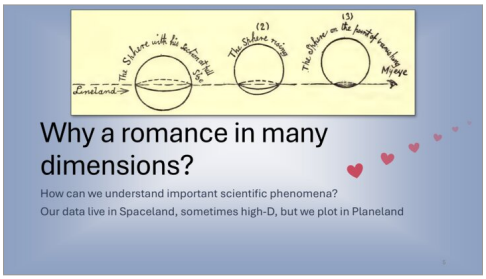
Roadmap: The book follows a linear path: plots of data, then models with one response, then models with many responses.

Today, I want to talk about the main themes of the book.

Themes:

The ellipse is the recurring visual device I use to understand multivariate phenomena.

I'll try to show how a wide variety of multivariate problems can be understood with insightful visual tools.



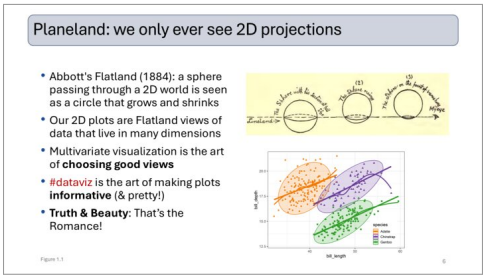
5. Why a romance in many dimensions?

I chose this subtitle to reflect the love I've always had about understanding high-D phenonema data through the visual and analysis tools we have at our disposal

Our goal in data analysis & visualization should always be to help understand scientific phenomena

Yet our data ofen live in Spaceland, sometimes high-D.

But, we are forced to plot them in 2D space.

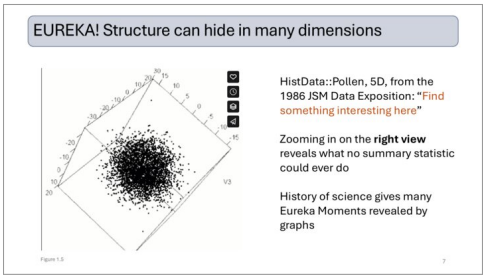


6. Planeland: we only ever see 2D projections

The book's subtitle is a nod to Abbott's 'Flatland: A Romance of Many Dimensions'.

The Flatlander sees only a slice of the sphere.

We are in the same position with p > 2 variables.



7. EUREKA! Structure can hide in many dimensions

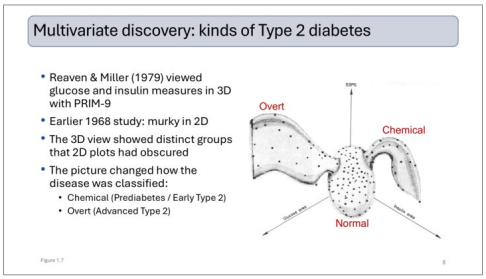
Animation (plays in slide show mode).

The Pollen dataset contains 5 variables measured on grains of pollen. It was created for the 1986 JSM Data Exposition:

The invitation was "Show your stuff!", "Find something interesting here".

It was a challenge, because, to most graphic methods of the day, it looks like a featureless cloud of points.

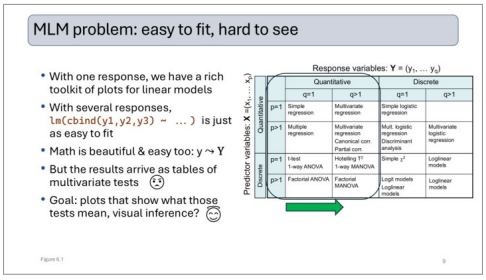
The secret hidden there was only revealed from one orientation in 5D space when you zoom in on the dense center, where the points spell out a word.



8. Multivariate discovery: kinds of Type 2 diabetes

An example of a real scientific discovery that came from looking at data in more than two dimensions.

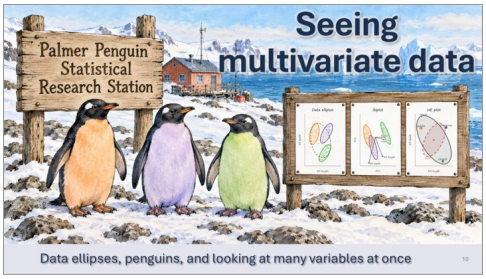
The Diabetes data come back later in the dimension reduction section.



9. MLM problem: easy to fit, hard to see

The table classifies linear model techniques by the number and type of responses and predictors. The right-hand column, multivariate responses, is where graphical methods have been thin.

The beauty of R formula notation is that it makes it easy to fit MLMs, just by adding additional responses to the LHS



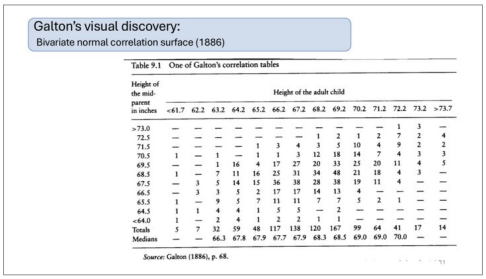
10. Seeing multivariate data

The first act in understanding multivariate data is to be able to "see" it

How can we do that?

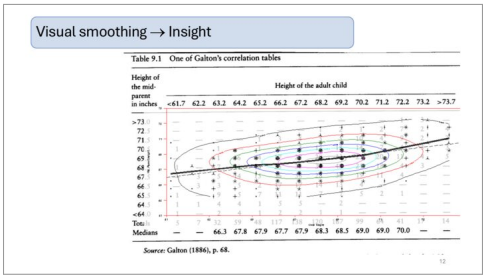
A primary visual device is an ellipsoid, used to represent multivariate location & scatter

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11. Galton's visual discovery: Bivariate normal correlation surface (1886)

My path to multivariate visualization begins with Galton's remarkable visual insight from a table of heights of parents and their offspring, that later gave rise to the theory of correlation and regression.



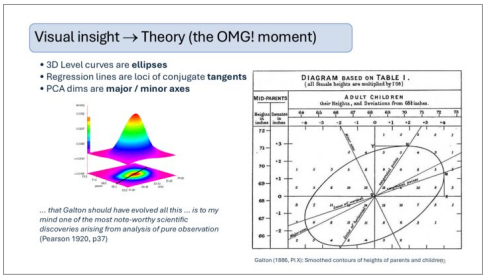
12. Visual smoothing → Insight

It is really more a semi-graphic table, recording frequency in this bivariate distribution.

Galton drew smoothed level curves- by eye and brain- of equal frequency

He noticed that these formed "concentric and similar ellipses," and

that the lines of means of x|y and y|x were the loci of conjugate tangents of the ellipses.



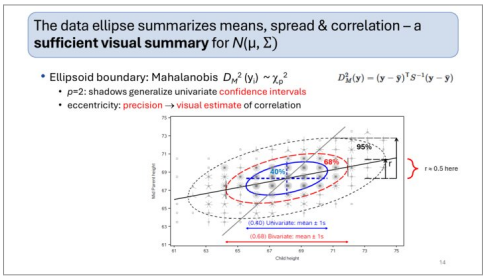
13. Visual insight → Theory (the OMG! moment)

From this, evolved the famous Galton diagram of correlation and regression shown at the right, where:

level curves of equal frequency are concentric ellipses and

the two regression lines, x|y and y|x are the loci of conjugate tangents.

Equally remarkable, he shows the major and minor axes of the ellipse, which would later be shown by Pearson to be the principal component directions in a bivariate distribution.

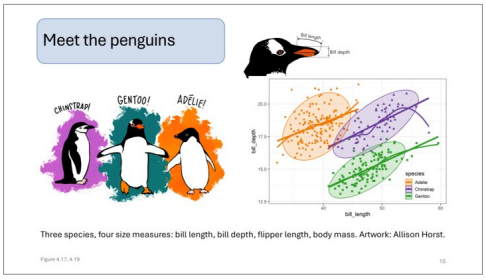


14. The data ellipse summarizes means, spread & correlation – a sufficient visual summary for N(μ, Σ)

Galton's data on the heights of parents and children. Data ellipses of 40%, 68% and 95% coverage.

The ellipse is the workhorse of the book:

the same idea returns as confidence ellipses for coefficients and as H and E ellipses for multivariate tests.

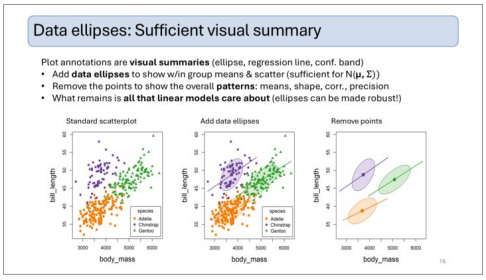


15. Meet the penguins

The Palmer penguins data is a main running example in the book

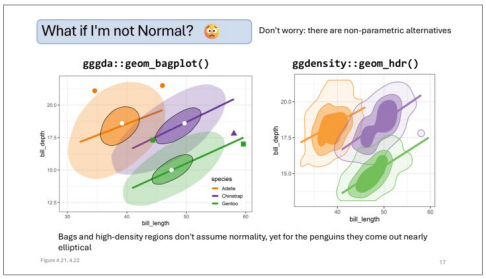
(The classic iris data seems to have been eclipsed b/c of Fisher's association with the eugenics movement because often called "Fisher's Iris data" even though it came from Edgar Anderson.)

Here: bill length and bill depth with data ellipses and regression lines for each species.



16. Data ellipses: Sufficient visual summary

To illustrate what I mean when I say “sufficient visual summary”– let’s start with an ordinary scatterplot…



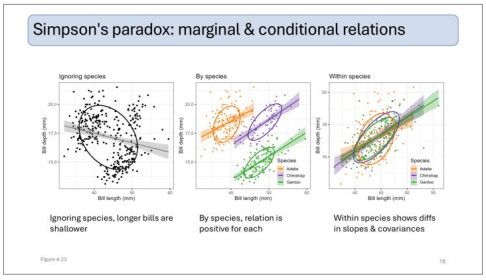
17. What if I’m not Normal?

If you’re worried about the assumption of bivariate normality in these plots, there are several non-parametric alternatives, just for the asking.

Each is a useful visual summary under relaxed conditions

Bagplot: robust generalization of a boxplot

High density regions



18. Simpson’s paradox: marginal & conditional relations

Data ellipses for the bill variables provide a beautiful example of Simpson’s paradox, and how data ellipses make that story clear

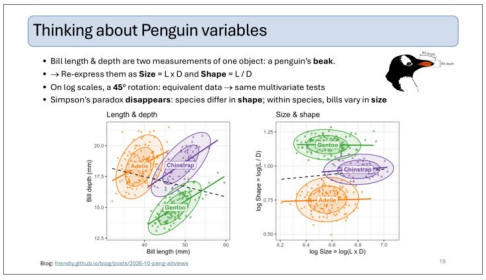
Panels:

(a) ignoring species,

(b) by species,

(c) pooled within species.

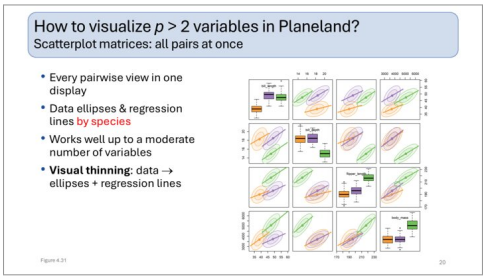
The data ellipses & regression lines make the reversal visually immediate.



19. Thinking about Penguin variables

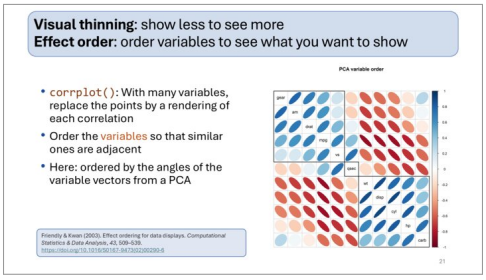
Data comes to us as variables were measured: Height and weight, where maybe BMI is more important to your doctor.

Thinking about what those variables really mean can help understanding



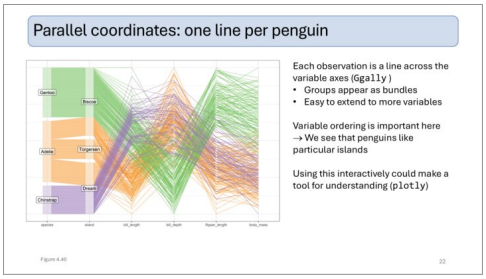
20. How to visualize p > 2 variables in Planeland? Scatterplot matrices: all pairs at once

Scatterplot matrix of the quantitative penguin variables.



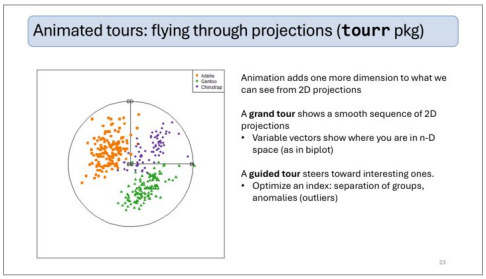
21. Visual thinning: show less to see more Effect order: order variables to see what you want to show

Corrplot of the crime data, with variables reordered. The ordering idea is taken up again under PCA.



22. Parallel coordinates: one line per penguin

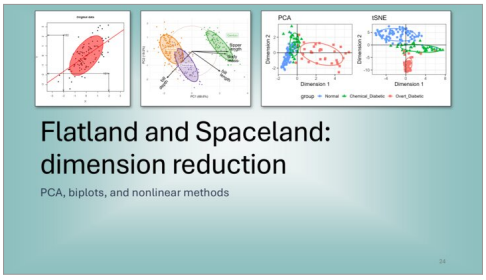
[Do this quick] Parallel coordinates plot of the penguin size variables, colored by species.



23. Animated tours: flying through projections (tourr pkg)

Yet another idea is to animate a sequence of 2D projections, on a travel through nD space

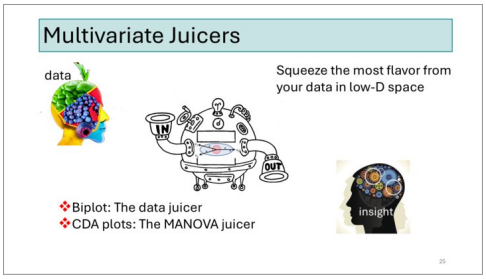
Animation (plays in slide show mode): grand tour of the penguin data. Guided tours optimize an index, e.g. separation of groups or the presence of anomalies.



24. Flatland and Spaceland: dimension reduction

Now I'd like to turn to another main theme of the book:

How dimension reduction techniques can help us to find the right views of multivariate data



25.

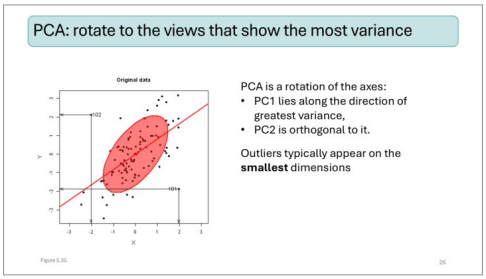
One key to this is to consider the wide class of these methods as "Multivariate Juicers"

I know this is SO cheesy!

But I find this a nice analogy to explain what we do with low-D methods

If we're plotting in 2D space, we want the view with the most flavor in the juice

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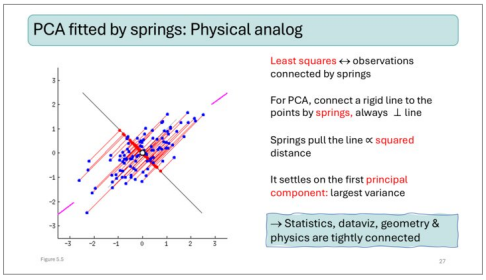
26. PCA: rotate to the views that show the most variance

Animation (plays in slide show mode): rotation from data space to PCA space, with two added points (101, 102).

They are not extreme on either original variable, but stand out on the smallest principal component.

Geometry of PCA as a rotation.

Tie back to the data ellipse: its major and minor axes are PC1 and PC2.

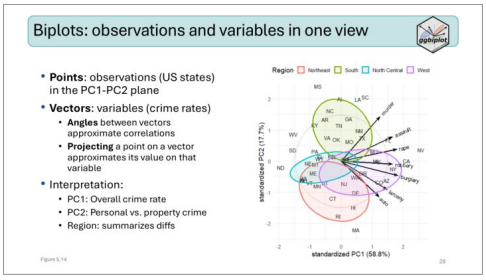


27. PCA fitted by springs: Physical analog

Animation (plays in slide show mode).

The blue points are connected to their projections on the red line by springs perpendicular to that line. A physical intuition for least squares.

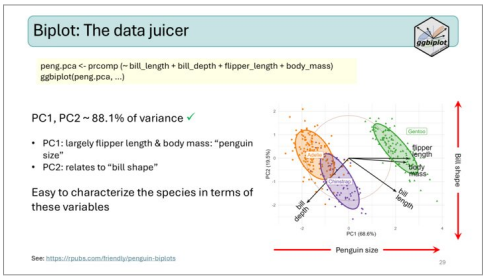
The point of this is that Statistics, dataviz, geometry & physics are often tightly connected. We can learn from one, using ideas of another



28. Biplots: observations and variables in one view

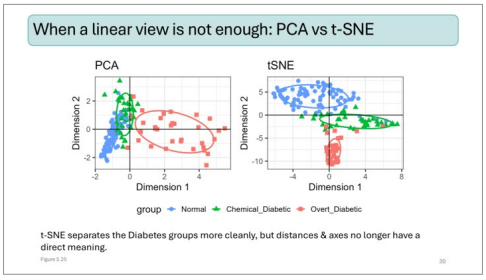
Basic biplot of the crime data.

The biplot is the low-D view used again and again: for collinearity, for canonical discriminant analysis, and for ridge regression.



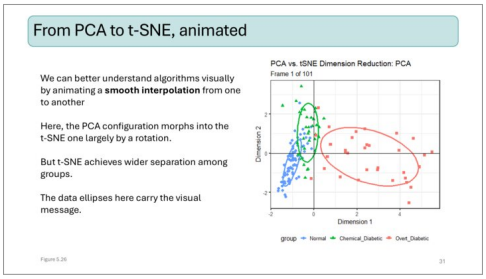
29. Biplot: The data juicer

Here's the biplot for the penguin data



30. When a linear view is not enough: PCA vs t-SNE

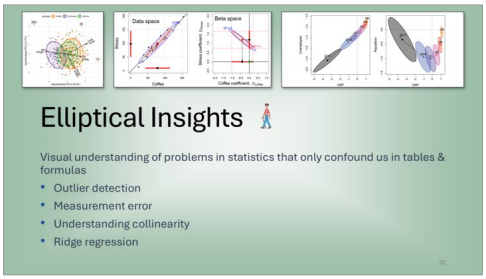
Comparison of the PCA and t-SNE 2D representations of the Diabetes data. Nonlinear methods preserve local neighborhoods at the cost of global geometry.



31. From PCA to t-SNE, animated

[Optional]

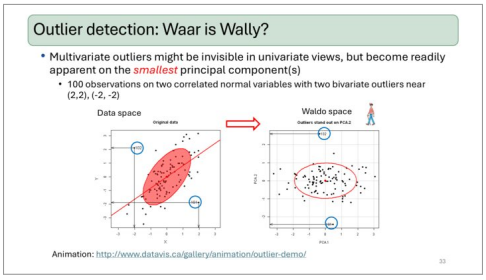
Animation (plays in slide show mode): interpolating between the PCA and t-SNE solutions for the Diabetes data.



32. Elliptical Insights

Another topic for today is how these ideas relate to what we do in statistical practice – analysing data, understanding the models we fit, and dealing with murky problems we can only see in tables…

I'll try to explain a few of these in terms of graphical methods based on ellipsoids

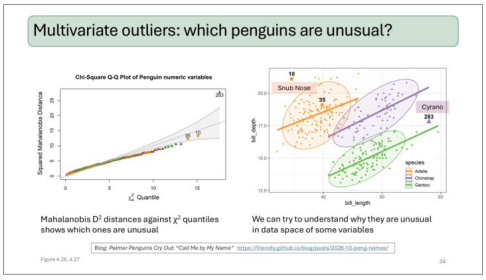


33. Outlier detection: Waar is Wally?

Like Galton, I've been impressed with the insights that can be had from elliptical thinking.

Just a simple demonstration of how outliers become apparent when viewed in the space of small principal components.

I call this Waldo space, by analogy with Martin Hansford's popular books, Where's Waldo? (Waar is Wally? in Dutch)



34. Multivariate outliers: which penguins are unusual?

Standard method: Chi-square QQ plot for the penguin data,

We can look at these in 2D plots – the unusual birds identified in the bill length-bill depth plot.

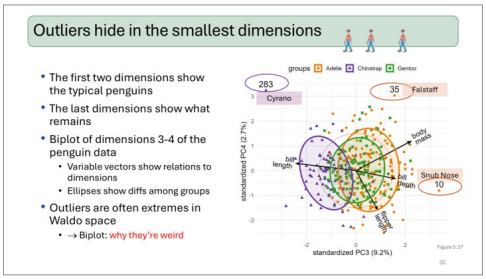
Two of them, nicknamed

Cyrano – really long billed Chinstrap

Snub Nose – stubbiest bill (short and deep)

Bird 35 doesn't seem at all unusual for an Adelie

I'll follow these birds in the influence and robust sections.



35. Outliers hide in the smallest dimensions

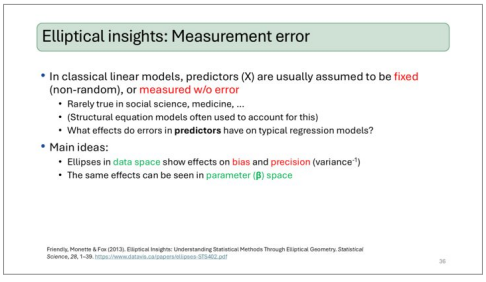
As I noted multivariate outliers often show up in the dimensions with the least variance.

In the biplot of dimensions 3-4 of the penguin data, variable vectors show how these dimensions relate to the data variables and also how outliers are unusual in the data

PC3: largely bill length (right to left) – Cyrano & Snub Nose are extremes

PC4: body mass vs flipper length – Falstaff is the stoutest bird

That's how they got their names!

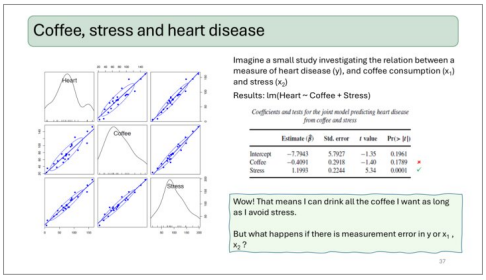


36. Elliptical insights: Measurement error

Another insight comes from considering the effect of measure error in predictors in linear models.

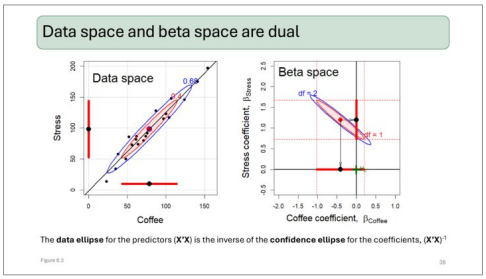
In data space, we can see these effects on bias and precision of coefficients.

But there is also a dual space of the parameters – beta space.



37. Coffee, stress and heart disease

(no notes)



38. Data space and beta space are dual

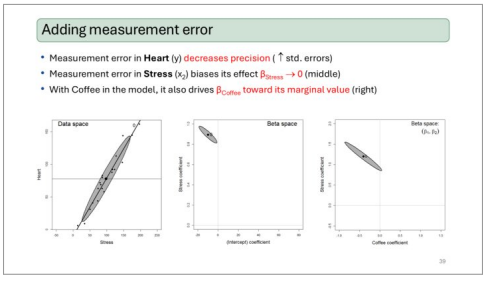
Data space and beta space for the coffee data.

Where the predictors have a wide spread, the coefficients are estimated precisely, and vice versa.

This shows why stress is significant in the full model, but coffee is not

Beautiful relations between statistics, geometry & matrix algebra

(This sets up collinearity.)



39. Adding measurement error

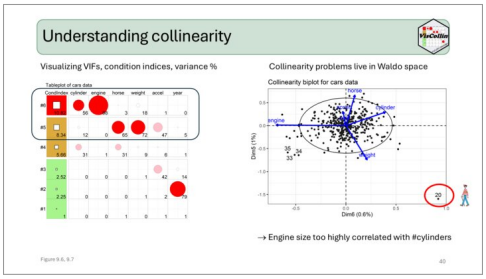
Animations (play in slide show mode), with increasing measurement error added to Stress.

Left, data space: the data ellipse for Heart and Stress gets wider, and the slope flattens.

Middle, beta space for the simple regression: the coefficient for Stress is attenuated toward 0.

OK, we understand that now. But, at the same time, in the model with both predictors (right), the effect of coffee becomes larger: it moves toward its marginal value, the slope you get ignoring Stress.

Elliptical insight: increasing error in x2 drives the coefficient for x1 toward the marginal model. You can also see that the standard error of the Coffee coefficient decreases as error in Stress increases.



40. Understanding collinearity

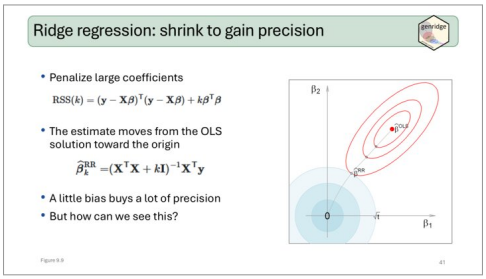
A visual approach also helps to understand problems of collinearity— high correlations among predictors— in linear models

The standard collinearity diagnostics table hides the important information.

A tableplot makes what you should attend to stand out– ordered by condition indices

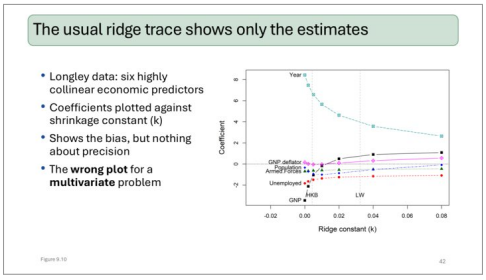
A collinearity biplot shows which variables are involved and possible outliers appear in Waldo space

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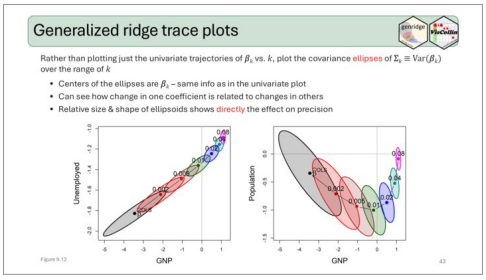
41. Ridge regression: shrink to gain precision

Geometric interpretation of ridge regression.



42. The usual ridge trace shows only the estimates

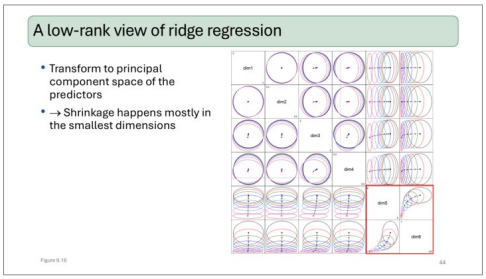
Univariate ridge trace plot for the Longley data.



43. Generalized ridge trace plots

Instead of plotting point estimates, we can see effects of ridge shrinkage by plotting bivariate ellipsoids for pairs of parameters

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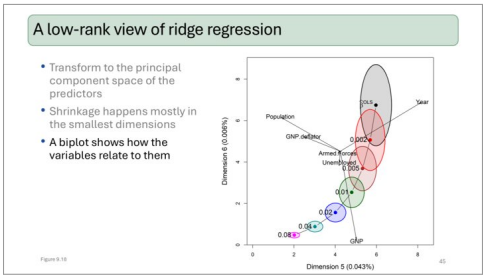


44. A low-rank view of ridge regression

Transforming the predictors to PCA space makes them uncorrelated.

Covariance ellipses for the dimensions show how shrinkage affects each pair.

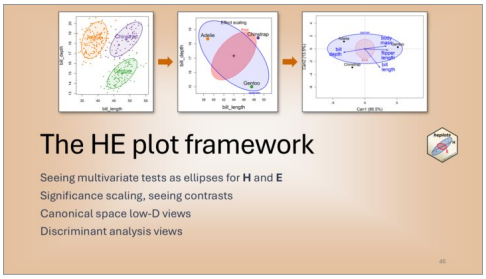
Nothing of interest except in the last two dimensions



45. A low-rank view of ridge regression

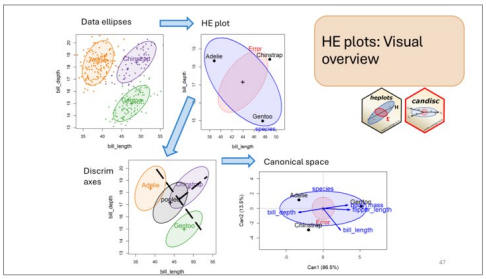
Transforming the predictors to PCA space makes them uncorrelated.

Biplot view of the ridge trace plot for the smallest dimensions shows clearly how precision is increased



46. The HE plot framework

Now, my favorite topic for today— how we can use these visual ideas to better understand the hypothesis tests we get in MLMs – MANOVA and MMRA



47. HE plots: Visual overview

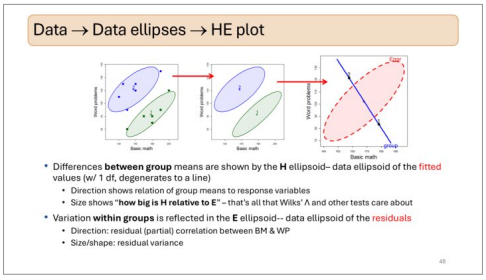
The HE plot framework starts with data ellipses for the groups in a multivariate linear model

An "HE plot" summarizes the evidence for differences among groups in a MANOVA design

Equally, this can be cast as a discriminant analysis problem

A simple description can be seen in an HE plot in canonical space

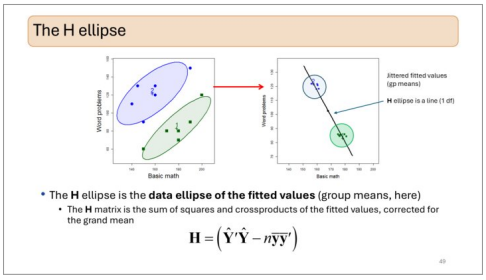
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48. Data → Data ellipses → HE plot

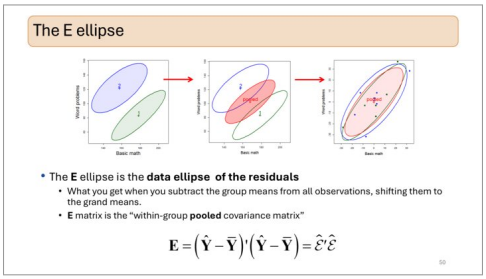
To simplify, I'll use Mathscore data: two groups, basic math (BM) and word problems (WP).

From the data to data ellipses to the HE plot.



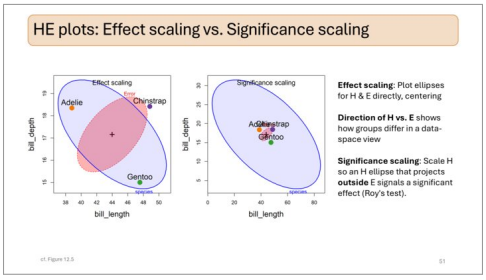
49. The H ellipse

The H ellipse is the data ellipse of the fitted values; with 1 df it collapses to a line.



50. The E ellipse

The E ellipse is the data ellipse of the residuals: the pooled within-group covariance.

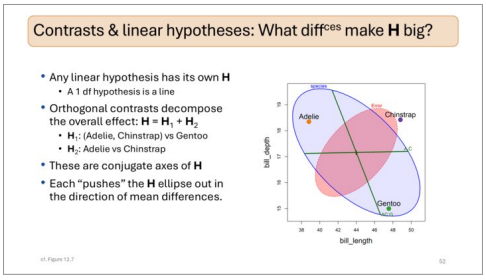


51. HE plots: Effect scaling vs. Significance scaling

HE plots for bill depth vs bill length in the penguin data, with effect scaling (left) and significance scaling (right).

E: within-group variation. H: variation of the group means.

The orientation of H shows how the groups differ. The species effect is so strong that, with significance scaling, H dwarfs E.

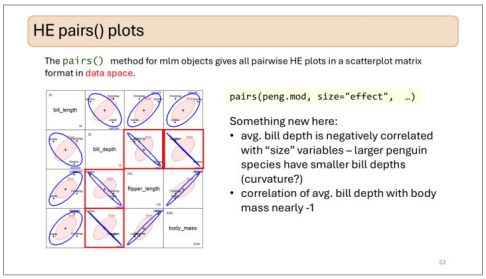


52. Contrasts & linear hypotheses: What diffces make H big?

HE plots also allow us to decompose an overall effect of group differences into effects related to contrasts among the factor levels

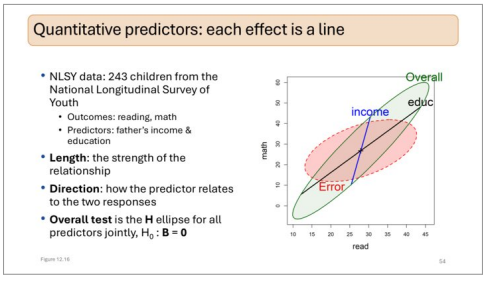
For bill length and depth in the penguin data, showing two orthogonal contrasts: Adelie vs Chinstrap (A:C), and those two vs Gentoo (AC:G).

Each is a line; together they are conjugate axes of the H ellipse.



53. HE pairs() plots

(no notes)



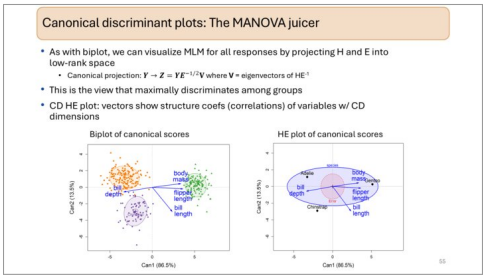
54. Quantitative predictors: each effect is a line

The same ideas are available for multivariate multiple regression.

HE plot for reading and math scores in the NLSY data, predicted from father's income and education.

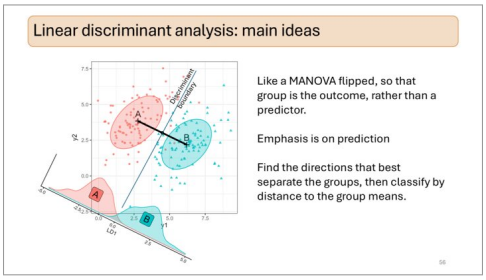
Each predictor has 1 df, so its H 'ellipse' collapses to a line: income relates mostly to math, education to both.

The green ellipse is the overall test that both coefficients are zero; the lines for the separate predictors always lie within it.



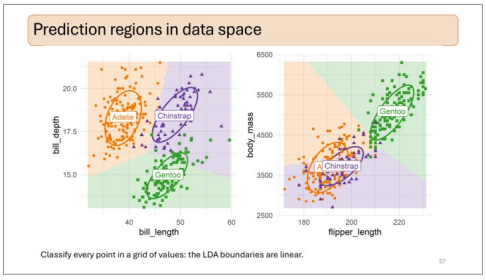
55. Canonical discriminant plots: The MANOVA juicer

(no notes)



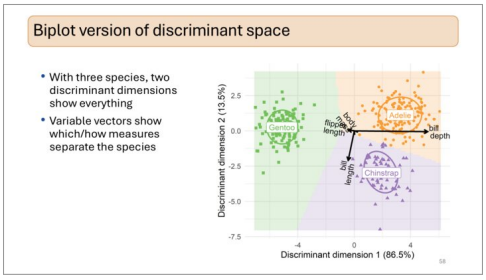
56. Linear discriminant analysis: main ideas

[OPTIONAL for a 45-minute talk] Illustration of the main ideas of linear discriminant analysis.



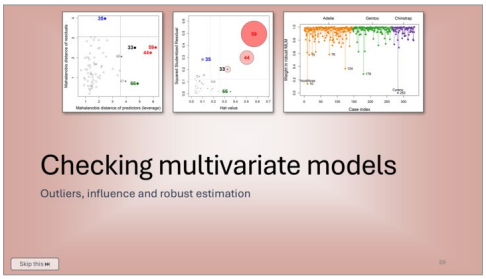
57. Prediction regions in data space

[OPTIONAL for a 45-minute talk] Linear discriminant prediction regions for the penguin data.



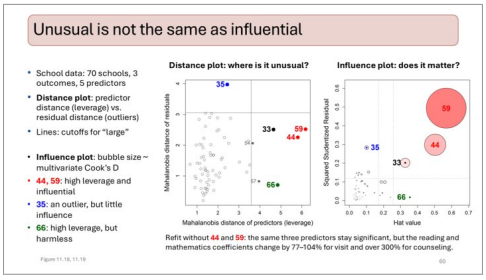
58. Biplot version of discriminant space

[OPTIONAL for a 45-minute talk] The penguin data in discriminant space, with variable vectors.



59. Checking multivariate models

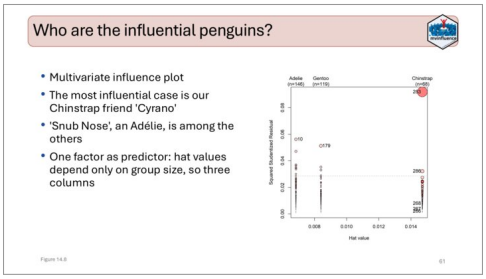
(no notes)



60. Unusual is not the same as influential

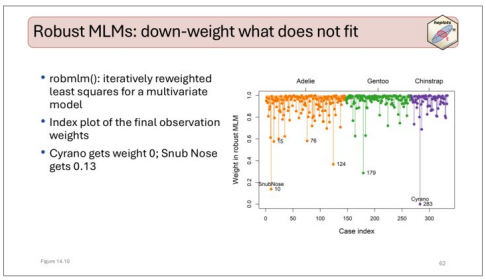
School data model: reading, mathematics and self-esteem on five predictors, 70 schools. Distance plot (heplots::distancePlot) and influence plot (mvinfluence::influencePlot), re-drawn as a matched pair with the same cases in the same colors (talks/R/school-diagnostics.R).

Click 1: influence plot. Click 2: sensitivity analysis. Cook's D: 59 = 2.99, 44 = 1.60, 33 = 0.71, 35 = 0.30, 66 = 0.07. Without 44 and 59, education, occupation and visit remain significant; relative change in coefficients: visit 104% (reading), 77% (math); counseling 332%, 305%.



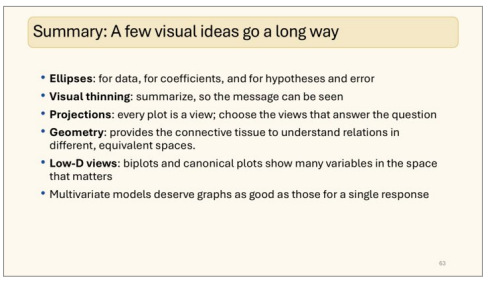
61. Who are the influential penguins?

Influence plot for the penguin MANOVA model.



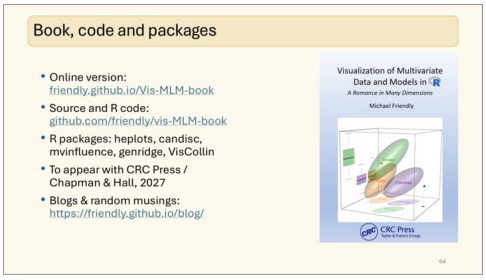
62. Robust MLMs: down-weight what does not fit

heplots::robmlm() applied to the penguin model. Most coefficients change little; the largest changes are for Chinstrap, reflecting the removal of Cyrano.



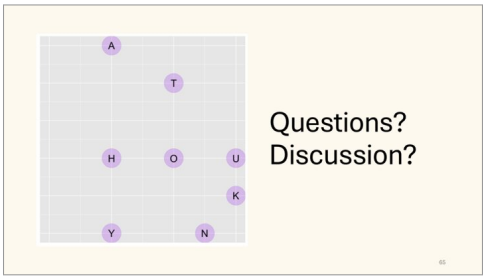
63. Summary: A few visual ideas go a long way

Summary of the themes that recur through the book.



64. Book, code and packages

Where to find the book and the software.



65. Questions? Discussion?

(no notes)
